TM Forum Component

Anomaly Management

TMFC041

Maturity Level: Beta	**Team Approved Date: 21-May-2026**
Release Status: Roadmap	**Approval Status: Not yet approved**
Version 1.3.0	**IPR Mode: RAND**

Notice

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TMFC041 – Anomaly Management

1. Overview

Component Name	ID	Description	ODA Function Block
Anomaly Management	TMFC041	The Anomaly Management component is responsible for predicting, detecting and managing anomalous patterns in the usage and behavior of products, services and resources in order to avoid occurrence of problems. Anomaly Management has functionality for predicting, detecting and flagging unusual patterns or outliers for products, services and resource entities, and initiating mitigatory actions to prevent outlier patterns in usage or behavior becoming problems. In the event of failure to predict or detect an early warning sign of a possible problem occurrence, the component hands off mitigatory actions to the Party Problem Management component.	Intelligence Management

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links

(no eTOM business activities are listed for this component in the YAML — see 2.2)

2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

(none listed in the component YAML)

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
Anomaly	Anomaly ABE represents any unusual or unexpected event, observation, pattern, or item that is significantly different from the norm that an enterprise wishes to detect, analyze, assess and mitigate within an autonomous environment. In order for the autonomous operations environment to behave within the calibrated operational level, anomalies may require administrative and operational guidance. Anomaly ABE entities provide a means of defining boundary conditions and lifecycle behavior for Anomalies, as well as providing control over the end-to-end management process for Anomalies.	Anomaly Entity	The Anomaly Entity ABE contains entities that are used to represent the various aspects of an Anomaly. This includes Product, Service, and Resource Anomalies as well as entities for concepts such as anomaly categorization, grouping, indicators, thresholds, consequences, reporting and notification.

Anomaly	Anomaly ABE represents any unusual or unexpected event, observation, pattern, or item that is significantly different from the norm that an enterprise wishes to detect, analyze, assess and mitigate within an autonomous environment. In order for the autonomous operations environment to behave within the calibrated operational level, anomalies may require administrative and operational guidance. Anomaly ABE entities provide a means of defining boundary conditions and lifecycle behavior for Anomalies, as well as providing control over the end-to-end management process for Anomalies.	Anomaly Management Plan	The Anomaly Management Plan ABE contains a group of entities that define the management plan for an Anomaly. For instance, the Anomaly management process typically involves stages such as Prediction, Detection, Mitigation, etc. Each stage necessitates a tailored strategy, which could include fixed logic for addressing the Anomaly and machine learning modules offering recommendations. The AnomalyManagementPlan ABE comprises the AnomalyManagementPlanSpecification and AnomalyManagementPlan entities. These entities, adhering to the EntitySpecification-Entity pattern, allow Anomaly Management implementers to specify additional attributes essential for articulating the management plan. The AnomalyManagementPlan is commonly utilized by the Closed Loop system, which oversees Anomalies, to configure the functions and workflows responsible for Anomaly Management.

Anomaly	Anomaly ABE represents any unusual or unexpected event, observation, pattern, or item that is significantly different from the norm that an enterprise wishes to detect, analyze, assess and mitigate within an autonomous environment. In order for the autonomous operations environment to behave within the calibrated operational level, anomalies may require administrative and operational guidance. Anomaly ABE entities provide a means of defining boundary conditions and lifecycle behavior for Anomalies, as well as providing control over the end-to-end management process for Anomalies.	Anomaly Specification	The Anomaly Specification ABE contains entities that are used to define the shared characteristics and behavior of Anomaly entities associated with Product, Service and Resource. It also includes related entities that specify the characteristic of grouping, categorization, indicators, and applicability of the Anomaly.

2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
1352	Anomaly Pattern Collection	Anomaly Management	Anomaly Detection	This function collects and stores "patterns" from various sources for use in anomaly detection. It takes input from a pattern discovery or pattern management source to be used internally for anomaly pattern processing
1353	Anomaly Normal Description Capture	Anomaly Management	Anomaly Detection	Anomaly Normal-Day Description Capture function is used to capture reference points to be used in the anomaly pattern processing. It takes as inputs of "normal-day" behavior, trend or pattern etc. and outputs a categorization of patterns

1354	Anomaly Pattern Correlation	Anomaly Management	Anomaly Detection	This function exposes analysis of the correlation between different anomaly patterns and their possible causes and impacts. It takes as input a pattern dataset and outputs the correlations between different anomaly patterns
1355	Anomaly Learning	Anomaly Management	Anomaly Detection	This function uses training methods to train and update the anomaly detection models based on data patterns and feedback (man/machine). It takes as input a dataset and outputs a model that can identify anomalies
1356	Anomaly Alerting	Anomaly Management	Anomaly Detection	This function sends alerts out (to users or other components) when an anomaly is detected. It takes as input the data and the model from the Anomaly Learning function, and outputs a signal to alert when the model identifies anomalies
1357	Anomaly Reporting	Anomaly Management	Anomaly Detection	This function generates and renders reports on the detected anomalies and their details. It takes as input the data and the anomalies identified, and outputs a report detailing these anomalies

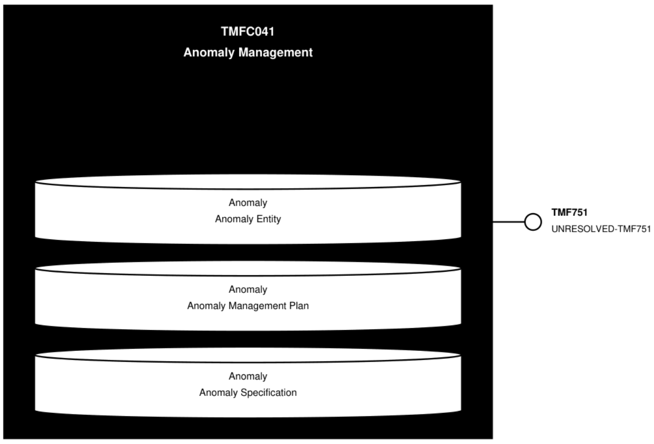
1358	Anomaly Pattern Analysis	Anomaly Management	Anomaly Detection	This function performs statistical and graphical analysis on patterns received and inferred to identify trends and outliers. It takes as input patterns and outputs an analysis of the patterns found to be outliers
1359	Univariate Anomaly Detection	Anomaly Management	Anomaly Detection	Detects anomalies in one variable, like revenue, cost, etc. The model is selected automatically based on pattern. It takes as input a univariate dataset and outputs the identified anomalies
1360	Multivariate Anomaly Detection	Anomaly Management	Anomaly Detection	Detects anomalies in multiple variables with correlations, which are usually gathered from products, services or resources (including devices, IoT and equipment etc.) or other complex systems. It takes as input a multivariate dataset and outputs the identified anomalies
1361	Anomaly Model Selection	Anomaly Management	Anomaly Detection	This function selects the best-fitting anomaly detection model from own or an external model repository for a given data set or scenario. It takes as input a dataset and potential models, and outputs the best model for a given anomaly detection requirement

1362	Real-time Model Inferencing	Anomaly Management	Anomaly Detection	This function (a.k.a Streaming Detection) performs anomaly detection on real-time data streams using a given or selected model. It takes as input new data and the selected model, and outputs identified anomalies in real-time
1363	Anomaly Batch Inferencing	Anomaly Management	Anomaly Detection	This function performs anomaly detection on batch data sets using given or selected model. It takes as input a batch of new data and the selected model, and outputs identified anomalies
1364	Anomaly Pattern Matching	Anomaly Management	Anomaly Detection	This function compares the detected anomalies with known anomaly patterns to find similarities and differences. It will compare observed patterns to known patterns (either anomalous or non-anomalous) to identify matches. It takes input from a dataset and a set of known patterns, and output matches between the observed and known patterns

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

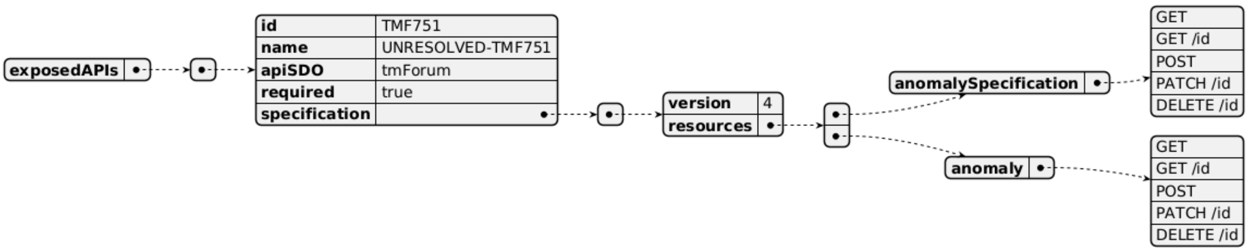
3.1. API Context Diagram



(SVG source: [TMFC041_API_Context.svg]() — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the `component-specification-documentation` skill's `scripts/render_api_context_svg.py`.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF751	UNRESOLVED-TMF751	Mandatory	4	anomalySpecification	GET, GET /id, POST, PATCH /id, DELETE /id
TMF751	UNRESOLVED-TMF751	Mandatory	4	anomaly	GET, GET /id, POST, PATCH /id, DELETE /id



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations

3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

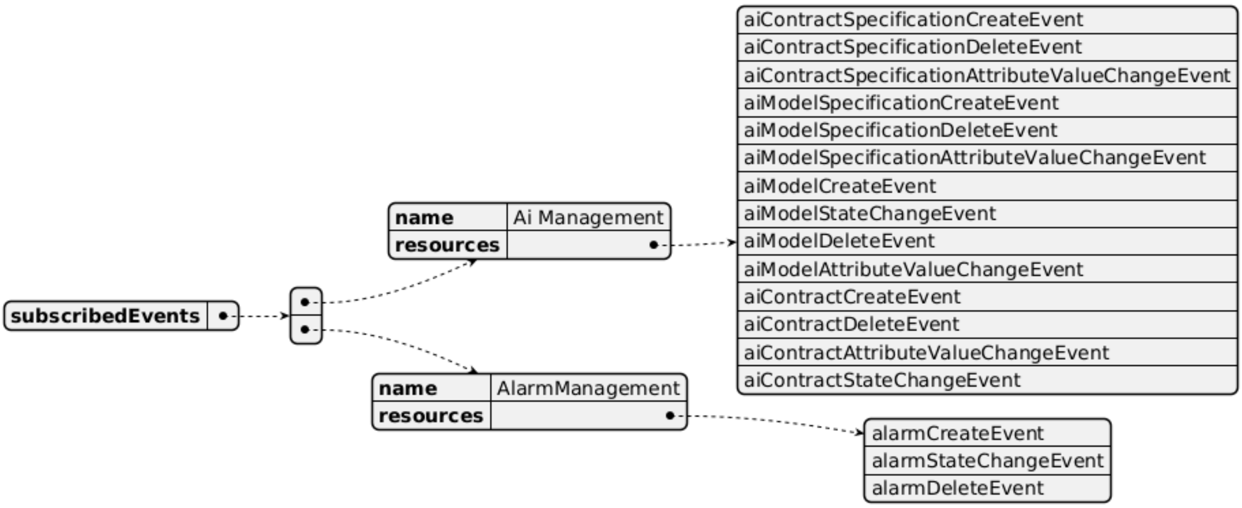
Published Events

API ID	API Name	Event Resources
TMF751	UNRESOLVED-TMF751	anomalyCreateEvent anomalyDeleteEvent anomalyAttributeValueChangeEvent anomalyStateChangeEvent anomalySpecificationStatusChangeEvent anomalySpecificationAttributeValueChangeEvent anomalySpecificationCreateEvent anomalySpecificationDeleteEvent

Subscribed Events

API ID	API Name	Event Resources

TMF915	AI Management API	aiContractSpecificationCreateEvent aiContractSpecificationDeleteEvent aiContractSpecificationAttributeValueChangeEvent aiModelSpecificationCreateEvent aiModelSpecificationDeleteEvent aiModelSpecificationAttributeValueChangeEvent aiModelCreateEvent aiModelStateChangeEvent aiModelDeleteEvent aiModelAttributeValueChangeEvent aiContractCreateEvent aiContractDeleteEvent aiContractAttributeValueChangeEvent aiContractStateChangeEvent
TMF642	Alarm Management API	alarmCreateEvent alarmStateChangeEvent alarmDeleteEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	*(none)*
SID	v25.0
Functional Framework	v24.0

5.2. Jira References

(none currently listed)

5.3. Further resources

1. IG1219 AI Closed Loop Automation - Anomaly Detection and Resolution
2. TR308 Anomaly Detector ODA Component Requirements v1.0.0 – TM Forum
3. TR309 Anomaly Predictor ODA Component Requirements v1.0.0 (tmforum.org)
4. TR310 Anomaly Mitigator ODA Component Requirements v1.0.0 (tmforum.org)
5. TR308A Anomaly Detector Business Process Scenarios v1.0.0 – TM Forum
6. TR309A Anomaly Predictor Business Process Scenarios v1.0.0 – TM Forum
7. TR310A Anomaly Mitigator Business Process Scenarios v1.0.0 – TM Forum
8. TR308E: Anomaly Detector Use Cases (WIP)
9. TMFSNNN: Anomaly Management - Product Ordering Anomaly Detection Detailed Use Case (WIP)
10. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0 Alpha	12-Aug-2024	Emmanuel A. Otchere, Manoj Nair	Initial Release

1.0.0 Alpha	21-Oct-2024	Manoj Nair	Removed TMF672 User & Roles Management from the dependent API list as it is expected to be supported by Canvas. Added the relevant resources and operations against each API to align with the component spec template
1.1.0	19-Nov-2024	Gaetano Biancardi	TMF688 removed from the core specification, moved to supporting functions. API version, only major version to be specified
1.1.0	26-Nov-2024	Amaia White	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	12-Aug-2024	Emmanuel A. Otchere, Manoj Nair	Component description, reference ISA objects, associated APIs and events
Pre-production	14-Oct-2024	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	26-Nov-2024	Amaia White	Updated to version 1.1.0
Pre-production	17-Feb-2025	Adrienne Walcott	Updated to Member Evaluated status

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team.

Team Member	Company	Role
Emmanuel A. Otchere	Huawei	Project Chair (AI-CLA), Author
Manoj Nair	Netcracker	Author
Gaetano Biancardi	Accenture	Additional Input

